

Asymptote

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1 Generalities

```
settings.tex = "lualatex";
settings.outformat = "pdf";
settings.prc = false;

texpreable("
\usepackage{fontspec}
\usepackage{amsmath}
\usepackage{amsthm}
\usepackage{unicode-math}
\setmainfont{Noto Sans}
\setsansfont{Noto Sans}
\setmathfont[Path=fonts/,
Extension=.ttf,
]{NotoSansMath-Regular}
");

unitsize(2cm);
pair A, B, C, D;
A=(0,0);
B=(2,0);
C=(4,0);
D=(4,2);
draw(A--B);
draw(C--D);
```



Figure 1: asy-1

```
size(5cm,0);  
pair A, B, C, D;  
A=(0,0);  
B=(2,0);  
C=(4,0);  
D=(4,2);  
draw(A--B);  
draw(C--D);
```

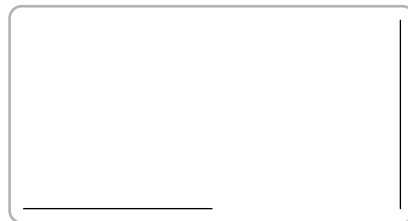


Figure 2: asy-2

```
size(10cm,0);  
pair A, B, C, D;  
A=(0,0);  
B=(2,0);  
C=(4,0);  
D=(4,2);  
draw(A--B);  
draw(C--D);
```

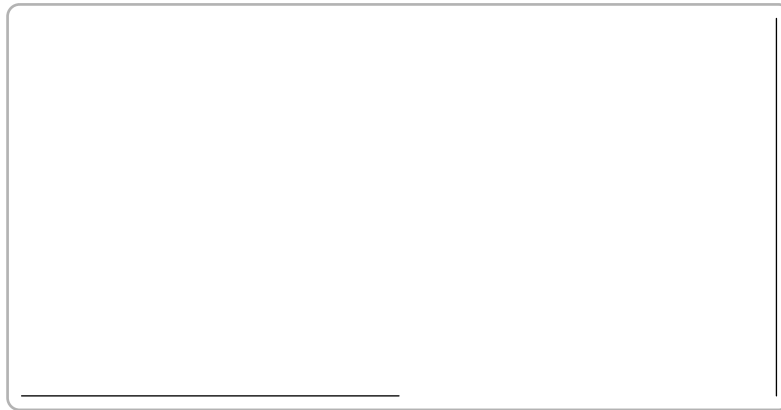


Figure 3: asy-3

```
size(5cm, 3cm);  
pair A, B, C, D;  
A=(0,0);  
B=(2,0);  
C=(4,0);  
D=(4,2);  
draw(A--B);  
draw(C--D);
```



Figure 4: asy-4

```
size(10cm, 2.5cm);  
pair A, B, C, D;  
A=(0,0);  
B=(2,0);  
C=(4,0);  
D=(4,2);  
draw(A--B);  
draw(C--D);
```



Figure 5: asy-5

```
size(10cm, 2.5cm, false); // width, height, and whether to preserve
// aspect ratio
pair A, B, C, D;
A=(0,0);
B=(2,0);
C=(4,0);
D=(4,2);
draw(A--B);
draw(C--D);
```



Figure 6: asy-6

```
size(3cm, 10cm, false);
pair A, B, C, D;
A=(0,0);
B=(2,0);
C=(4,0);
D=(4,2);
draw(A--B);
draw(C--D);
```

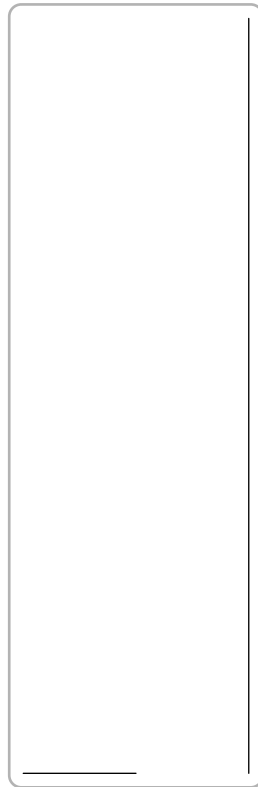


Figure 7: asy-7

```
pair A, B, C, D;  
A=(0,0);  
B=(3cm,0);  
C=(6cm,0);  
D=(6cm,4cm);  
draw(A--B);  
draw(C--D);
```



Figure 8: asy-8

```
unitsize(1cm);  
pair A, B, C, D;  
A=(0,0);
```

```
B=(3,0);  
C=(6,0);  
D=(6,4);  
draw(A--B);  
draw(C--D);
```



Figure 9: asy-9

```
unitsize(x=2cm, y=1cm);  
pair A, B, C, D;  
A=(0,0);  
B=(3,0);  
C=(6,0);  
D=(6,4);  
draw(A--B);  
draw(C--D);
```



Figure 10: asy-10

```
size(10cm,0);  
draw(scale(4)*unitcircle);  
dot((0, 0));  
dot((4, 0));
```



```
fixedscaling((-8,-8),(6,6));
shipout(bbox(Fill(lightgrey))));
```

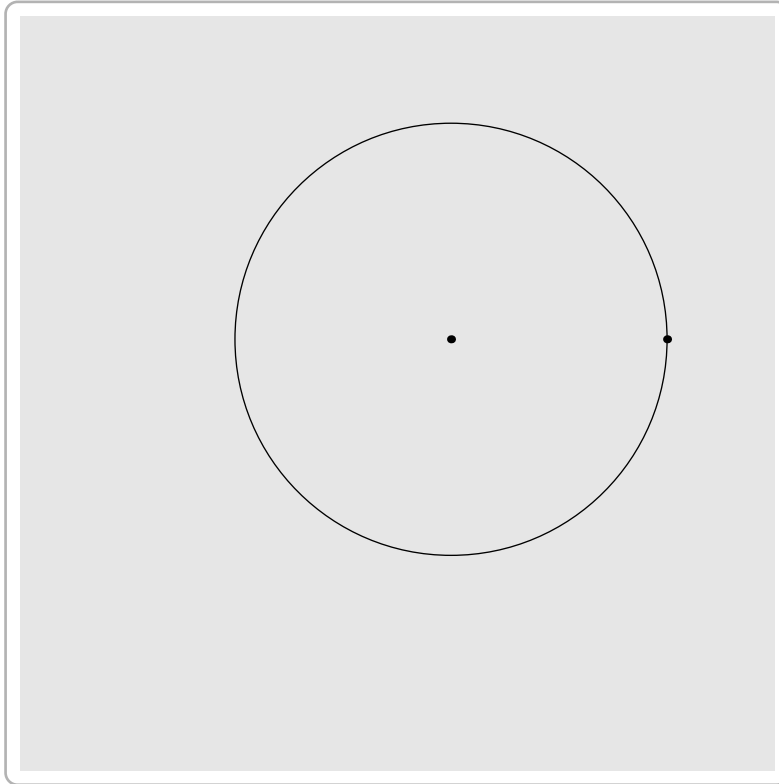


Figure 11: asy-11

```
size(10cm,0);
path p=(0,0)--(1,0);
frame object;
draw(object, scale(8cm)*p);
add(object);
add(object, (0,-40));
```

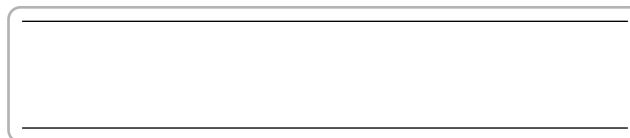


Figure 12: asy-12

```
size(10cm,0);
path p=(0,0)--(1,0);
picture object;
draw(object,scale(8cm)*p);
```

```
add(object);  
add(object,(0,-40)); // Adds truesize object to currentpicture
```



Figure 13: asy-13

```
size(6cm,6cm);  
pair A=(0,0), B=(1,0), C=(0,1);  
draw (A--B--C);
```

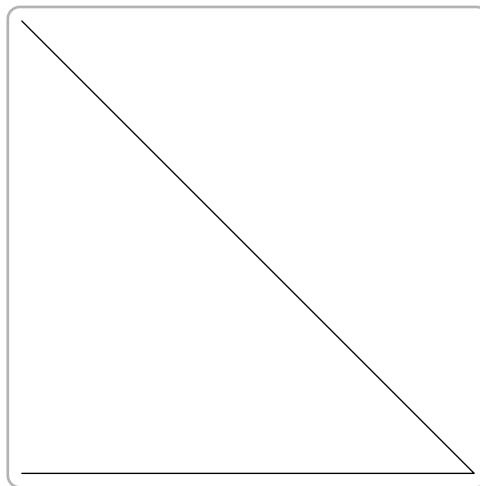


Figure 14: asy-14

```
size(6cm, 6cm);  
pair A=(0,0), B=(1,0), C=(0,1);  
draw (A--B--C--cycle);
```

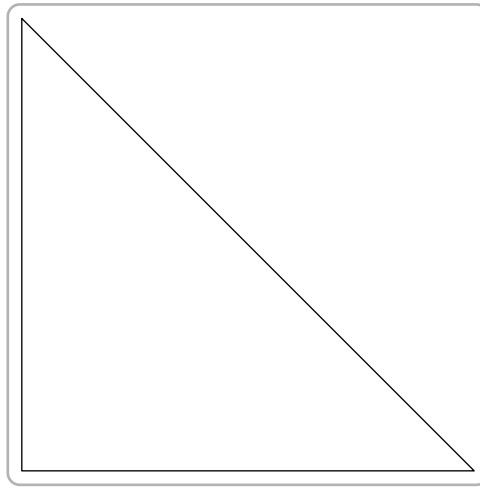


Figure 15: asy-15

```
size(6cm, 6cm);  
pair [] A;  
A[0]=(-1, -1);  
A[1]=( 1, -1);  
A[2]=( 1,  1);  
A[3]=(-1,  1);  
draw (A[0]--A[1]--A[2]--A[3]--cycle);  
draw (A[0]--A[2]);  
draw (A[1]--A[3]);
```

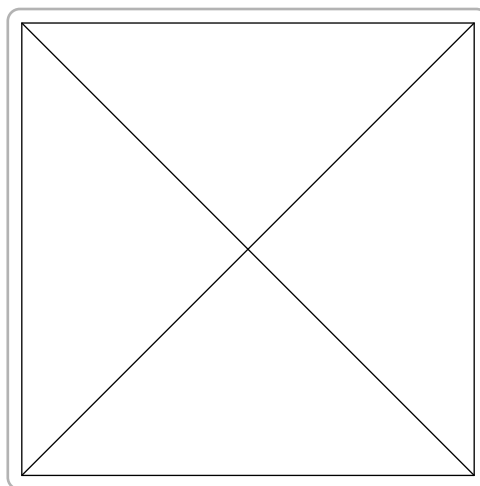


Figure 16: asy-16

```
size(6cm,6cm);  
draw(unitsquare);  
dot((0,0)--(1,1)--(0,1)--(1,0));
```

```
draw((0,0)--(1,1));
draw((1,0)--(0,1));
```

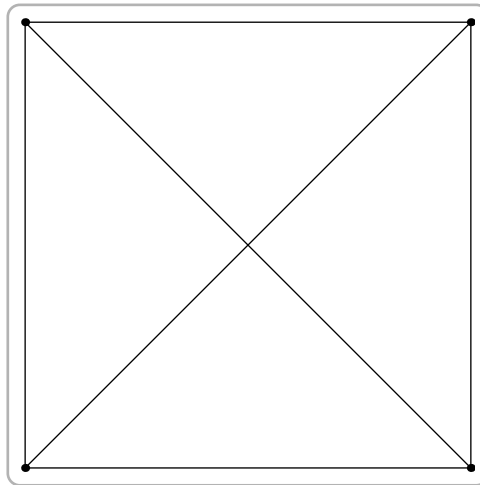


Figure 17: asy-17

```
size(6cm,3cm,false);
draw(unitsquare);
dot((0,0)--(1,1)--(0,1)--(1,0));
draw((0,0)--(1,1));
draw((1,0)--(0,1));
```

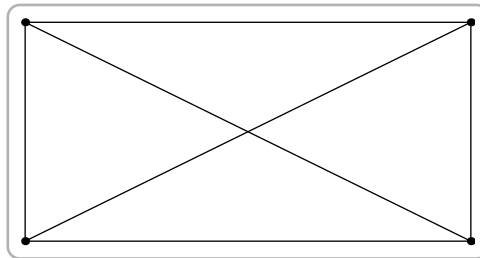


Figure 18: asy-18

```
size(8cm,0);
pair A,B,C;
A=(0,0);B=(1,0);C=(2,0);
draw(A,8bp+black);
draw(B,8bp+blue);
draw(C,linewidth(8bp));
```



Figure 19: asy-19

```

unitsize(3cm);
pair A=(0,0), B=(1,0), C=(0,1);
path trig;
trig=A--B--C--cycle;
draw(trig);
dot(trig,linewidth(8bp));
dot(shift(3,0)*trig,red+8bp);

```



Figure 20: asy-20

```

size(6cm,6cm);
pair A, B, C;
A=(0,0); B=(1,0); C=(0,1);
draw(A--B--C--cycle);
draw (midpoint(A--B) -- C);
draw (B+0.5*(C-B) -- A);
draw (interp(A,C,0.5) -- B);

```

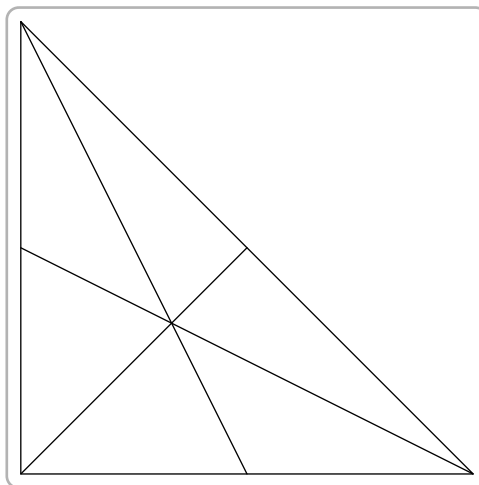


Figure 21: asy-21

```

size(6cm,6cm);
pair A, B, C;
A=(0,0); B=(1,0); C=(0,1);
draw(A--B--C--cycle);
draw (midpoint(A--B) -- C);
draw (B+0.5*(C-B) -- A);
draw (interp(A,C,0.5) -- B);
dot(A/3 + B/3 + C/3);

```

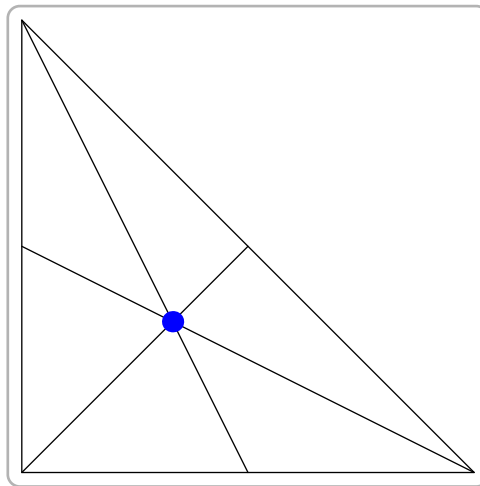


Figure 22: asy-22

```

size(6cm,0);
pair A=(0,0), B=(1,0), C=(0,1);
draw(A--B--C--cycle);
draw(A--B,linewidth(4bp));

```

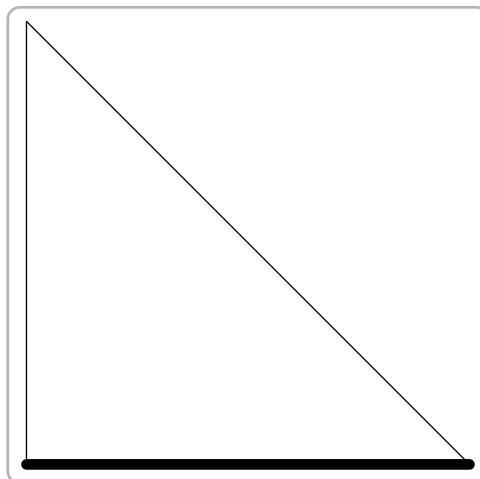


Figure 23: asy-23

```
size(6cm,0);
pair A=(0,0), B=(1,0), C=(0,1);
draw(A--B--C--cycle);
draw(A--B,(red+green));
```

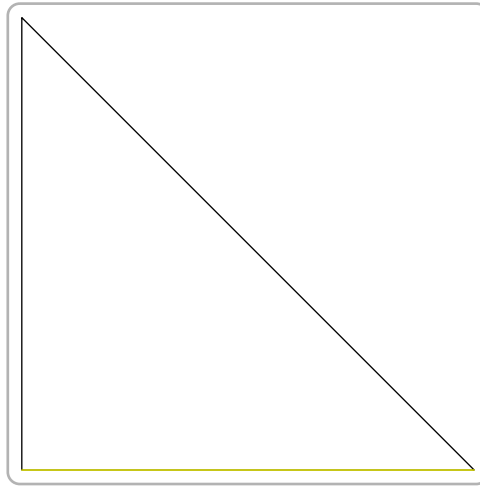


Figure 24: asy-24

```
size(6cm,0);
pair A=(0,0), B=(1,0), C=(0,1);
draw(A--B, 0.8white+6bp);
draw(A--C, 0.6white+6bp);
draw(B--C, 0.4white+6bp);
draw(A--B--C--cycle, yellow);
```

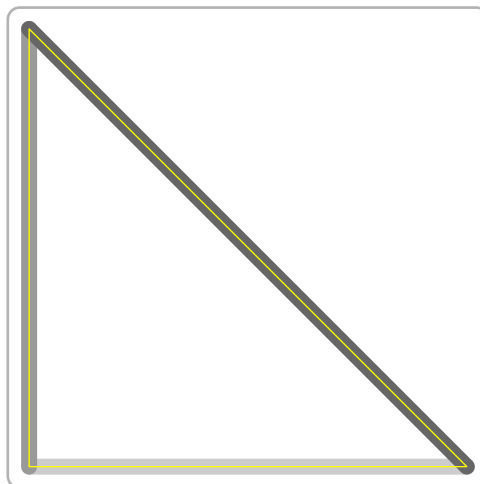


Figure 25: asy-25

```
size(8cm,0);
path p1=(0,0)--(1,1);
```

```

path p2=(2,0)--(3,1);
draw(p1,linewidth(8mm));
draw(p2,linewidth(8mm));
draw(p1--p2, 4mm+0.8green);
draw(reverse(p1)--p2, 2mm+.8red);

```

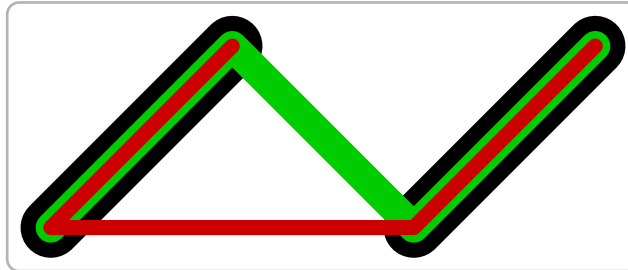


Figure 26: asy-26

```

size(8cm,0);
pair A=(0,0), B=(1,0);
dot("$A$",A,N);
dot("$B$",B,N);
draw(interp(A,B,-0.5)--interp(A,B,1.25));

```

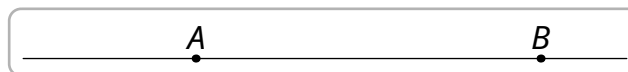


Figure 27: asy-27

```

unitsize(cm);
path line=(0,0)--(6,0);
draw(line,solid);
draw(shift(0,-1)*line,dotted);
draw(shift(0,-2)*line,dashed);
draw(shift(0,-3)*line,longdashed);
draw(shift(0,-4)*line,dashdotted);
draw(shift(0,-5)*line,longdashdotted);

```

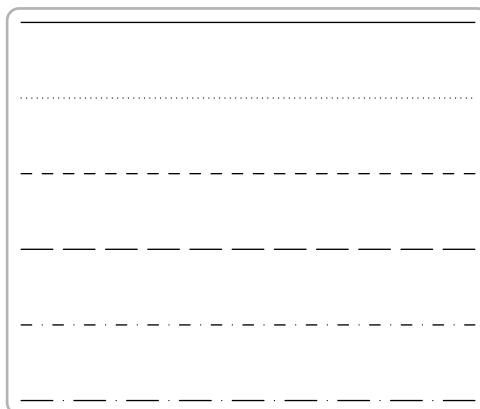



Figure 28: asy-28

```
unitsize(1cm);
path line=(0,0)--(6,0);
pen my_pen=linetype("24 8 8 8");
draw(line,linetype("24 8 8 8"));
draw(shift(0,-1)*line,my_pen+0.8red);
draw(shift(0,-2)*line,my_pen+0.7green+4bp);
draw(shift(0,-3)*line,linetype("24 8 8 8",false)+0.7blue+4bp);
draw(shift(0,-4)*line,linetype("24 8 8 8",true,false)+4bp);
draw(shift(0,-5)*line,linetype("24 8 8 8",false,false)+0.5red+blue+4bp);
```

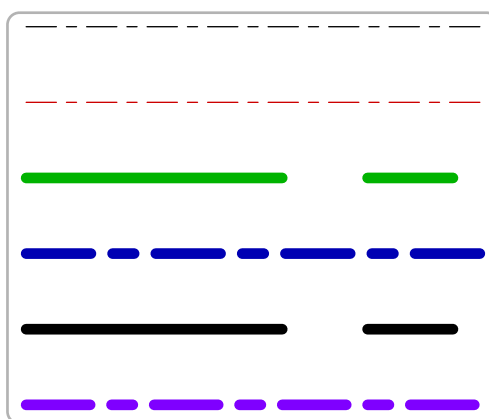


Figure 29: asy-29

```
size(8cm,0);
transform T=shift(0,-0.125);
path g=(0,0)--(1,0);
int i=-1;
draw(T^(++i)*g, Arrows());
draw(T^(++i)*g, 0.8*green, Arrows(SimpleHead));
```

```

draw(T^(++i)*g, 0.8*red, Arrows(SimpleHead,size=5mm));
draw(T^(++i)*g, Arrows(HookHead));
draw(T^(++i)*g, 0.8*green, Arrows(HookHead,size=5mm));
draw(T^(++i)*g, 0.8*red, Arrows(HookHead(barb=20),size=5mm));
draw(T^(++i)*g, 2mm+0.8*blue, Arrows(HookHead(dir=60),size=5mm, Fill));
draw(T^(++i)*g, 0.8*yellow, Arrows(HookHead(barb=-10,dir=40),size=5mm));
draw(T^(++i)*g, Arrows(TeXHead));
draw(T^(++i)*g, 0.8*green, Arrows(TeXHead,size=5bp));
draw(T^(++i)*g, 5bp+0.8*red, Arrows(TeXHead));

```

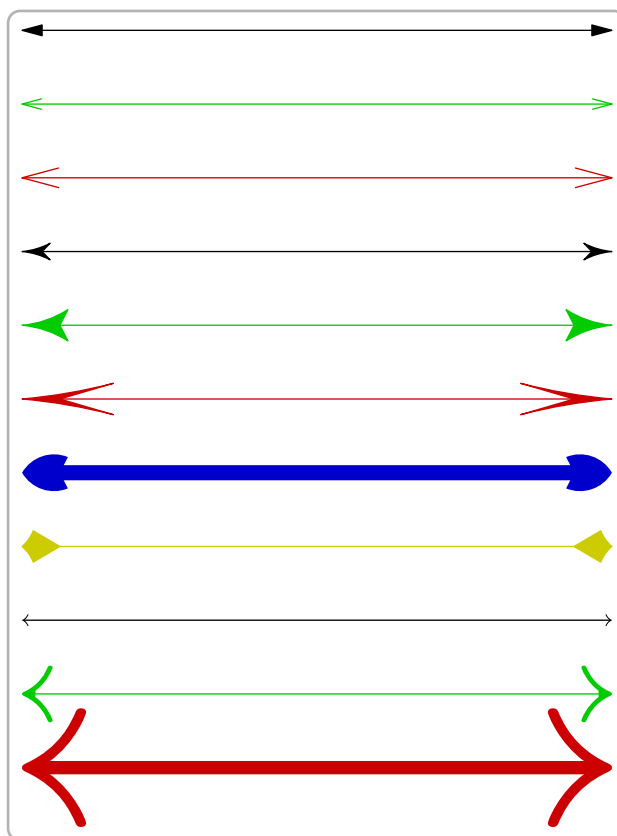


Figure 30: asy-30

```

path line=(0,0)--(5cm,0);
draw(line,Arrow);
draw(shift(0,-1cm)*line, BeginArrow);
draw(shift(0,-2cm)*line, MidArrow);
draw(shift(0,-3cm)*line, Arrows);
draw(shift(0,-4cm)*line, linewidth(3bp), Arrow(20bp));
draw(shift(0,-5cm)*line, linewidth(3bp), BeginArrow(20bp));
draw(shift(0,-6cm)*line, linewidth(3bp), MidArrow(20bp));
draw(shift(0,-7cm)*line, linewidth(3bp), Arrows(20bp));

```

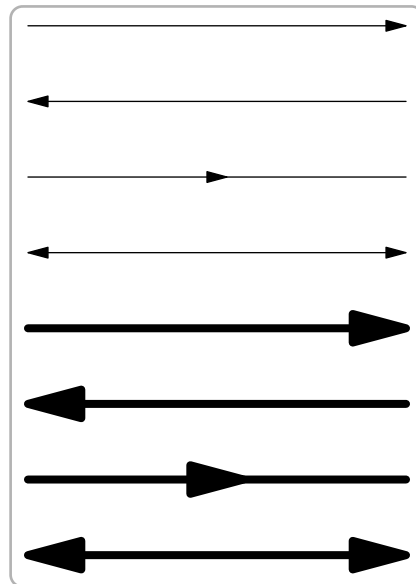


Figure 31: asy-31

```

path line=(0,0)--(5cm,0);
DefaultHead=HookHead;
draw(shift(6cm,0)*line,Arrow());
draw(shift(6cm,-1cm)*line, BeginArrow());
draw(shift(6cm,-2cm)*line, MidArrow());
draw(shift(6cm,-3cm)*line, Arrows());
draw(shift(6cm,-4cm)*line, linewidth(3bp), Arrow(20bp));
draw(shift(6cm,-5cm)*line, linewidth(3bp), BeginArrow(20bp));
draw(shift(6cm,-6cm)*line, linewidth(3bp), MidArrow(20bp));
draw(shift(6cm,-7cm)*line, linewidth(3bp), Arrows(20bp));

```

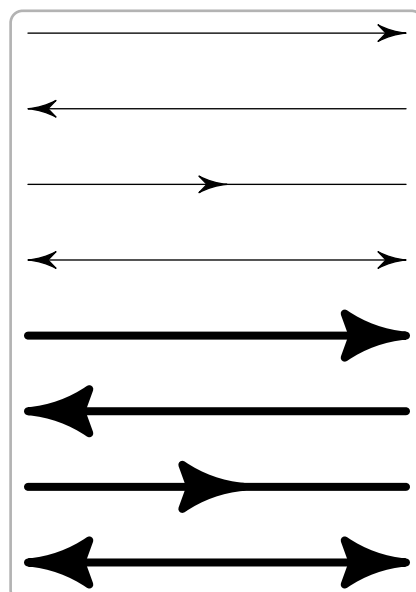


Figure 32: asy-32

```

path line=(0,0)--(5cm,0);
DefaultHead=SimpleHead;
draw(shift(6cm,0)*line,Arrow());
draw(shift(6cm,-1cm)*line, BeginArrow());
draw(shift(6cm,-2cm)*line, MidArrow());
draw(shift(6cm,-3cm)*line, Arrows());
draw(shift(6cm,-4cm)*line, linewidth(3bp), Arrow(20bp));
draw(shift(6cm,-5cm)*line, linewidth(3bp), BeginArrow(20bp));
draw(shift(6cm,-6cm)*line, linewidth(3bp), MidArrow(20bp));
draw(shift(6cm,-7cm)*line, linewidth(3bp), Arrows(20bp));

```

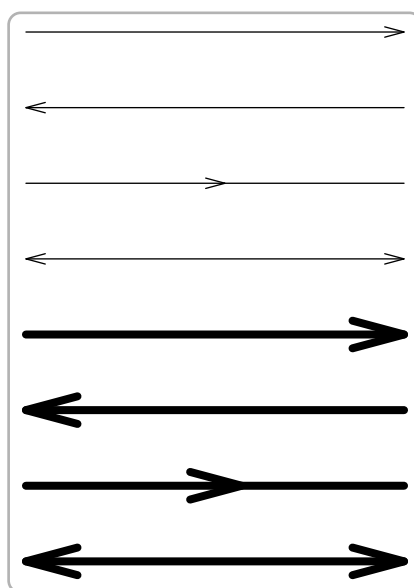


Figure 33: asy-33

```

path line=(0,0)--(5cm,0);
DefaultHead=TeXHead;
draw(shift(6cm,0)*line,Arrow());
draw(shift(6cm,-1cm)*line, BeginArrow());
draw(shift(6cm,-2cm)*line, MidArrow());
draw(shift(6cm,-3cm)*line, Arrows());
draw(shift(6cm,-4cm)*line, linewidth(bp), Arrow());
draw(shift(6cm,-5cm)*line, linewidth(2bp), BeginArrow(5bp));
draw(shift(6cm,-6cm)*line, linewidth(2bp), MidArrow(5bp));
draw(shift(6cm,-7cm)*line, linewidth(2bp), Arrows(5bp));

```

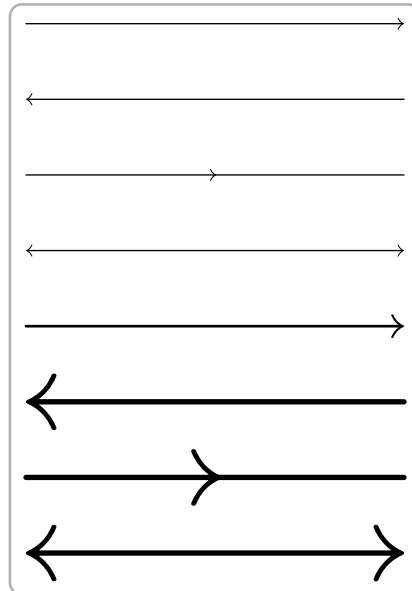


Figure 34: asy-34

```

path line=(0,0)..(2.5cm,2.5cm)..(5cm,0);
draw(line,ArcArrow);
draw(shift(0,-2cm)*line,BeginArcArrow);
draw(shift(0,-4cm)*line,MidArcArrow);
draw(shift(0,-6cm)*line,ArcArrows);
draw(shift(6cm,0)*line,3bp+0.8red,ArcArrow(20bp));
draw(shift(6cm,-2cm)*line,3bp+0.8red,BeginArcArrow(20bp));
draw(shift(6cm,-4cm)*line,3bp+0.8red,MidArcArrow(20bp));
draw(shift(6cm,-6cm)*line,3bp+0.8red,ArcArrows(20bp));

```

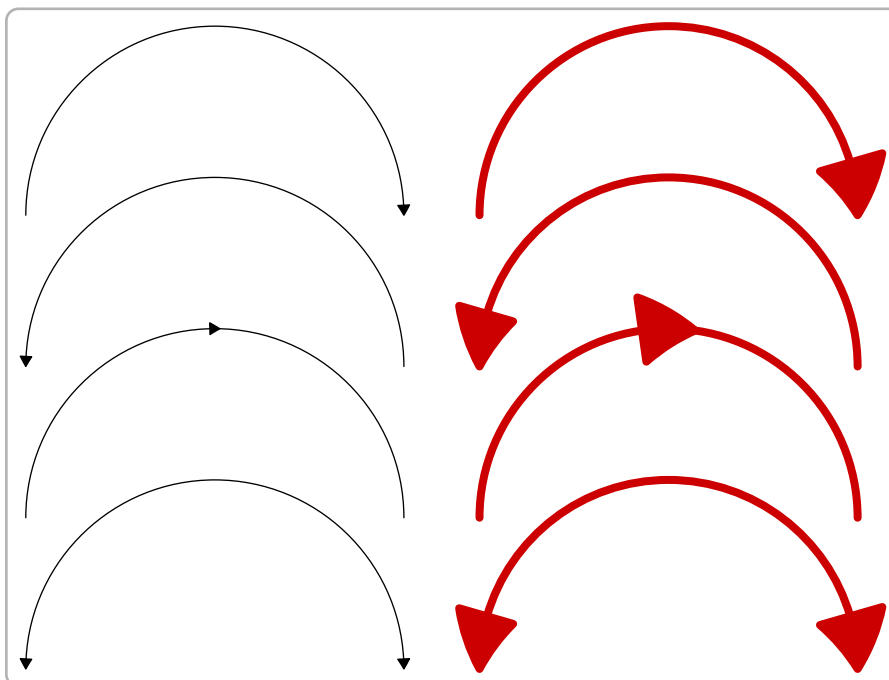


Figure 35: asy-35

```

DefaultHead=HookHead;
path line=(0,0)..(2.5cm,2.5cm)..(5cm,0);
draw(line,ArcArrow());
draw(shift(0,-2cm)*line,BeginArcArrow());
draw(shift(0,-4cm)*line,MidArcArrow());
draw(shift(0,-6cm)*line,ArcArrows());
draw(shift(6cm,0)*line,3bp+0.8red,ArcArrow(20bp));
draw(shift(6cm,-2cm)*line,3bp+0.8red,BeginArcArrow(20bp));
draw(shift(6cm,-4cm)*line,3bp+0.8red,MidArcArrow(20bp));
draw(shift(6cm,-6cm)*line,3bp+0.8red,ArcArrows(20bp));

```

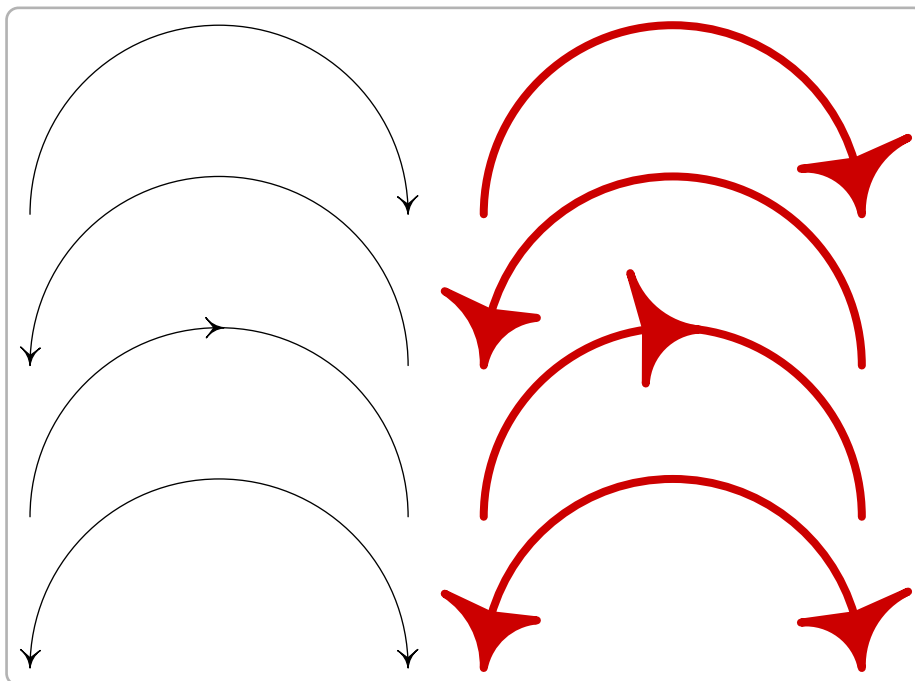


Figure 36: asy-36

```

DefaultHead=SimpleHead;
path line=(0,0)..(2.5cm,2.5cm)..(5cm,0);
draw(line,ArcArrow());
draw(shift(0,-2cm)*line,BeginArcArrow());
draw(shift(0,-4cm)*line,MidArcArrow());
draw(shift(0,-6cm)*line,ArcArrows());
draw(shift(6cm,0)*line,3bp+0.8red,ArcArrow(20bp));
draw(shift(6cm,-2cm)*line,3bp+0.8red,BeginArcArrow(20bp));
draw(shift(6cm,-4cm)*line,3bp+0.8red,MidArcArrow(20bp));
draw(shift(6cm,-6cm)*line,3bp+0.8red,ArcArrows(20bp));

```

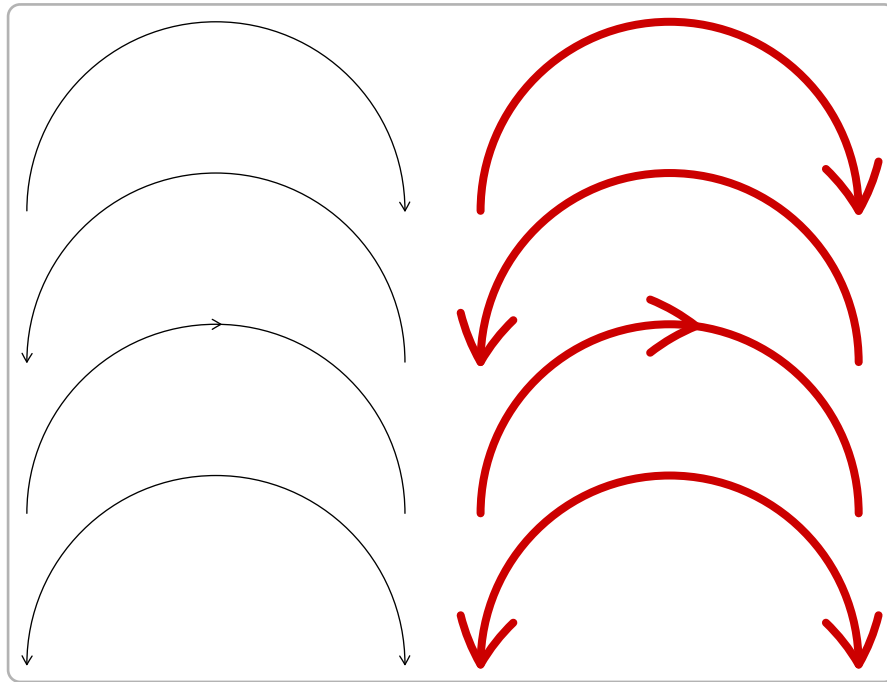


Figure 37: asy-37

```

DefaultHead=TeXHead;
path line=(0,0)..(2.5cm,2.5cm)..(5cm,0);
draw(line,Arrow());
draw(shift(0,-2cm)*line,BeginArrow());
draw(shift(0,-4cm)*line,MidArrow());
draw(shift(0,-6cm)*line,Arrows());
draw(shift(6cm,0)*line,3bp+0.8red,Arrow(5bp));
draw(shift(6cm,-2cm)*line,3bp+0.8red,BeginArrow(5bp));
draw(shift(6cm,-4cm)*line,3bp+0.8red,MidArrow(5bp));
draw(shift(6cm,-6cm)*line,3bp+0.8red,Arrows(5bp));

```

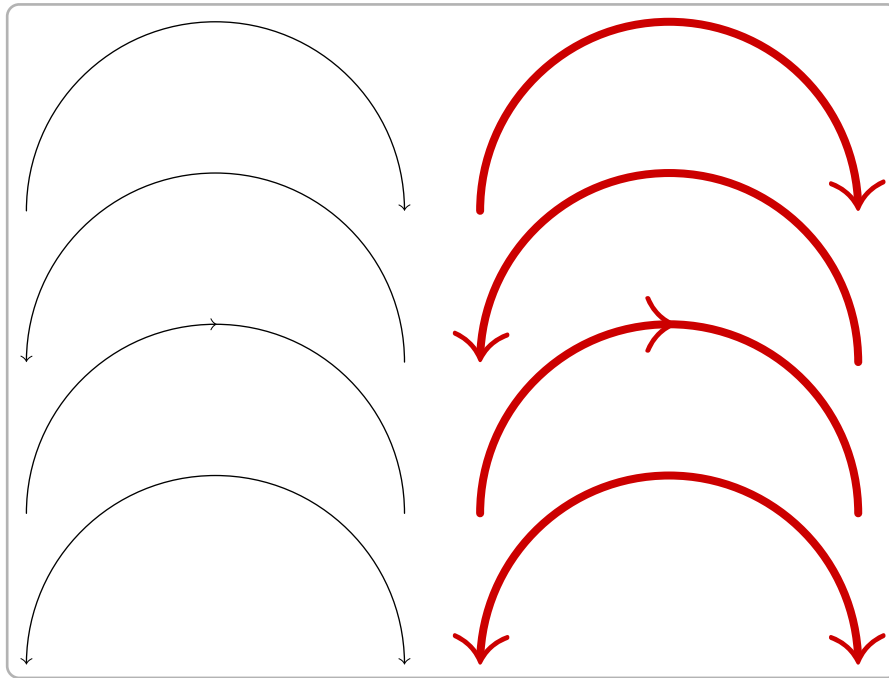


Figure 38: asy-38

```

path line=(0,0)--(5cm,0);
draw(line,Arrow(20bp,filltype=NoFill));
draw(shift(0,-1cm)*line,Arrow(20bp,filltype=FillDraw(0.8red)));
draw(shift(0,-2cm)*line,Arrow(20bp,filltype=Fill(0.8red)));
draw(shift(0,-3cm)*line,Arrows(filltype=FillDraw(-10,10,0.8red)));

```

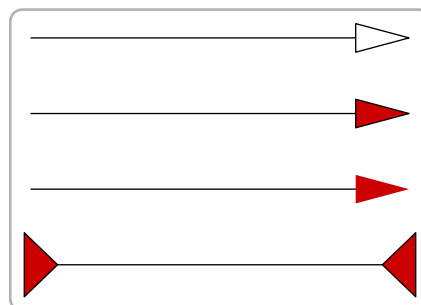


Figure 39: asy-39

```

DefaultHead=HookHead;
path line=(0,0)--(5cm,0);
draw(line,Arrow(20bp,filltype=NoFill));
draw(shift(0,-1cm)*line,Arrow(20bp,filltype=FillDraw(0.8red)));
draw(shift(0,-2cm)*line,Arrow(20bp,filltype=Fill(0.8red)));
draw(shift(0,-3cm)*line,Arrows(filltype=FillDraw(-10,10,.8red)));

```

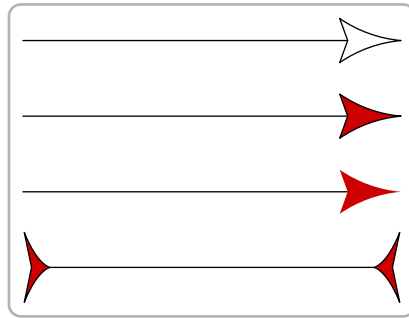



Figure 40: asy-40

```
DefaultHead=TeXHead;
path line=(0,0)--(5cm,0);
draw(line,Arrow(10bp,filltype=NoFill));
draw(shift(0,-2cm)*line,Arrow(10bp,filltype=FillDraw(0.8red)));
draw(shift(0,-4cm)*line,Arrow(10bp,filltype=Fill(0.8red)));
draw(shift(0,-6cm)*line,Arrows(filltype=FillDraw(-10,10,0.8red)));
```

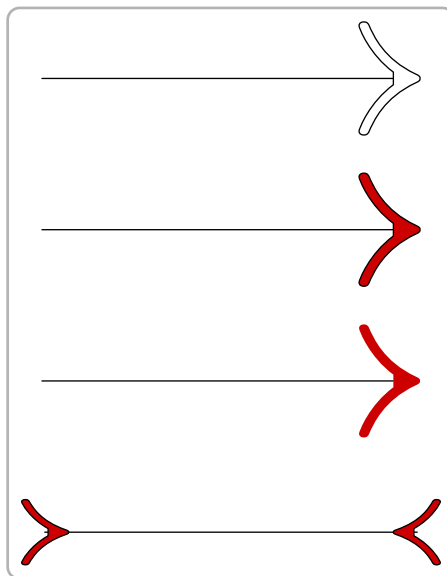


Figure 41: asy-41

```
path line=(0,0)--(5cm,0);
draw(line,linewidth(5bp),Arrow);
draw(shift(0,-2cm)*line,linewidth(5bp),Arrow(30bp,10));
draw(shift(0,-5cm)*line,linewidth(30bp),Arrow(20bp,50));
draw(shift(0,-8cm)*line,linewidth(10bp),Arrow(20bp,50,filltype=NoFill));
```

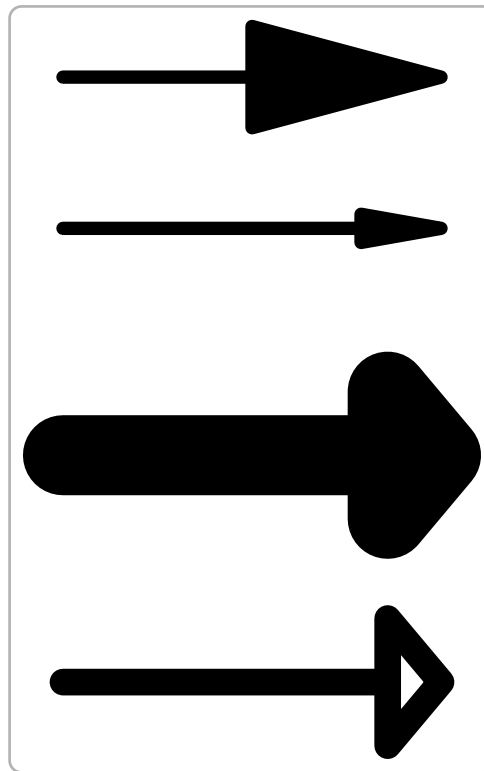


Figure 42: asy-42

```
size(5cm,0);
path line=(0,0)--(5,0);
draw(line,Arrow(20bp,position=0.75));
draw(shift(0,-2)*line,Arrow(20bp,40,0.75,filltype=NoFill));
position pos=BeginPoint;
pos.position=0.75;
draw(shift(0,-4)*line,BeginArrow(20bp,pos));
draw(shift(0,-6)*line,BeginArrow(20bp,40,pos,filltype=NoFill));
```

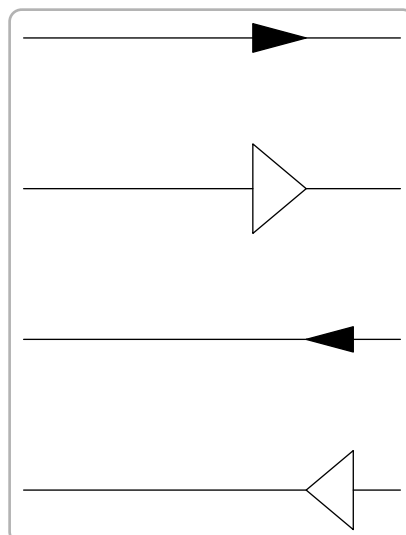


Figure 43: asy-43

```
unitsize(8cm);
path g=(0,0)--(1,0);
draw(g,Arrow(Relative(0.45)));
add(arrow(g,Relative(0.55)));
```



Figure 44: asy-44

```
size(8cm,0);
path g = (0,0)--(1,0);
draw(g,Arrow(Relative(0.75)));
add(arrow(reverse(g), invisible, FillDraw(green,red), Relative(0.75)));
```

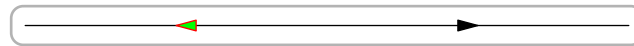


Figure 45: asy-45

```
size(5cm,0);
path line=(0,0)--(5,0);
transform T=shift(0,-1);
draw(line, Bar);
draw(T*line, BeginBar);
draw(T^2*line, Bars);
draw(T^3*line, Arrow, BeginBar);
```

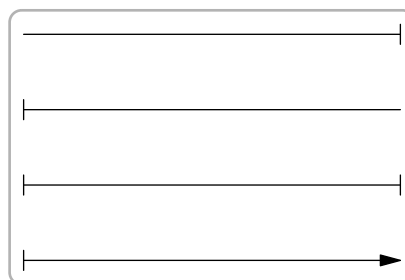


Figure 46: asy-46

```
path line=(0,0)--(5cm,0);
transform T=shift(0,-1cm);
draw(line,linewidth(1mm),Bars);
draw(T^2*line,Bars(5mm));
```

```
draw(T^3*line,linewidth(1mm),Bars(5mm));
draw(T^4*line,dotted+red,Bars);
```

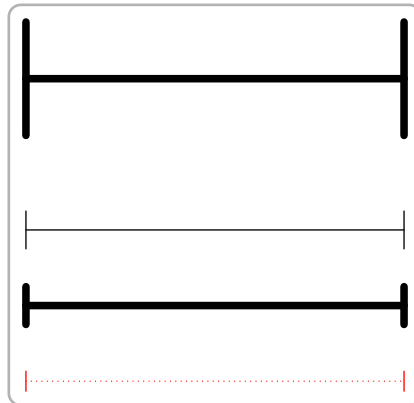


Figure 47: asy-47

```
size(6cm,0);
draw((-0.5,0)--(1,0));
draw((0,-0.5)--(0,1));
draw((0,0)--(0.75,0),linewidth(1mm),Arrow(3mm));
draw((0,0)--(0,0.75),linewidth(1mm),Arrow(3mm));
```

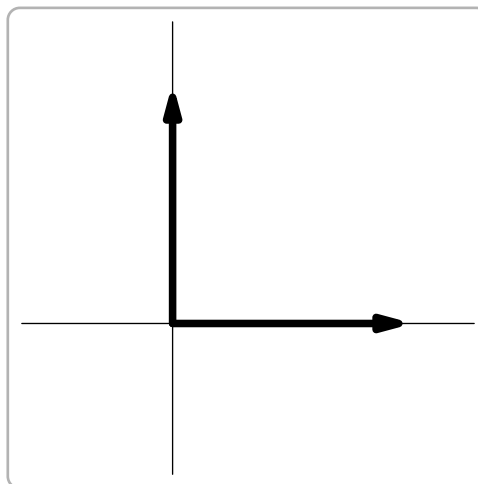


Figure 48: asy-48

```
fill((0,0)--(0,6cm)--(6cm,0)--cycle, gray);
```

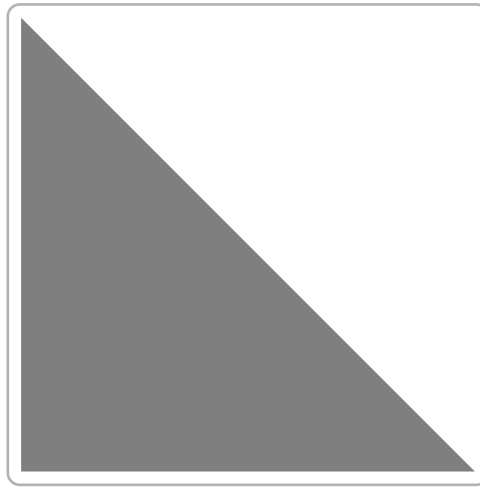


Figure 49: asy-49

```
filldraw((0,0)--(0,6cm)--(6cm,0)--cycle,gray);
```

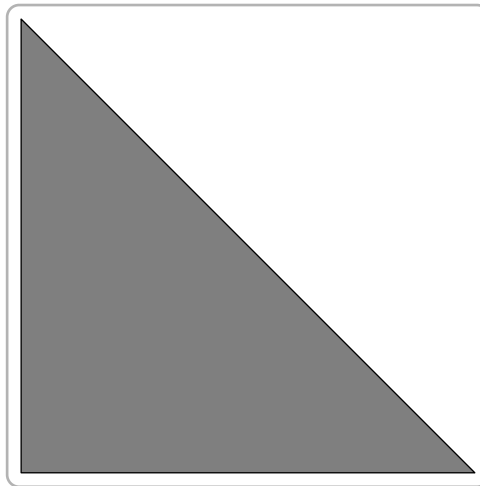


Figure 50: asy-50

```
filldraw((0,0)--(0,6cm)--(6cm,0)--cycle,  
fillpen=gray, drawpen=linewidth(1mm) + 0.8 * red);
```

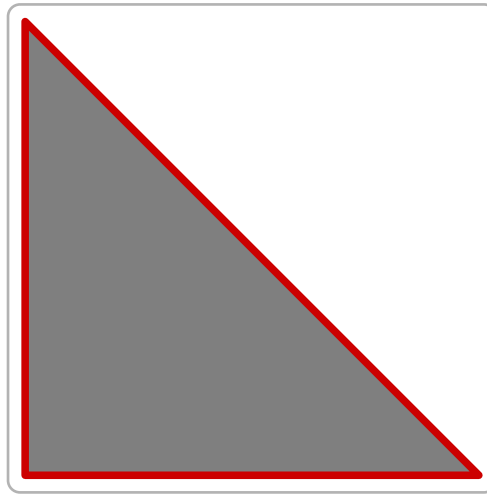


Figure 51: asy-51

```
pen[][] p={{rgb(black)}, {rgb(.8red)}}};  
latticeshade((0,0)--(0,6cm)--(6cm,0)--cycle, p);
```

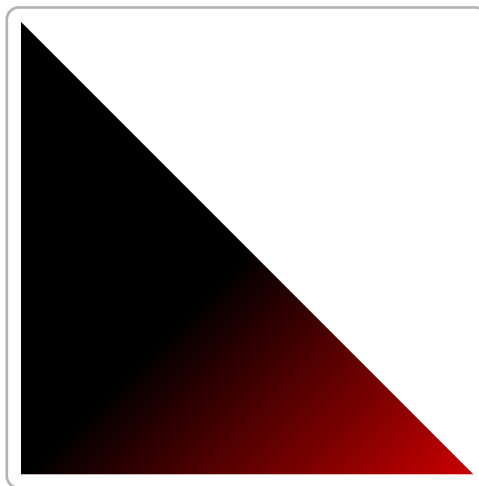


Figure 52: asy-52

```
pen[][] p={{rgb(black),rgb(black)}, {rgb(red),rgb(green)}}};  
latticeshade((0,0)--(0,6cm)--(6cm,0)--cycle,p);
```

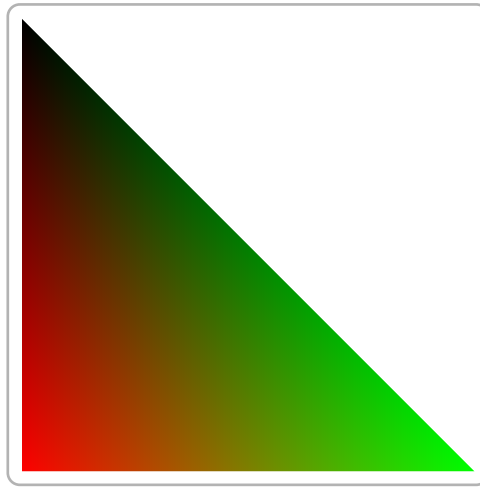


Figure 53: asy-53

```
size(8cm,0);
pen[][] p={{rgb(white),rgb(grey),rgb(black)},
  {red,green,blue},
  {cyan,magenta,yellow}};
latticeshade(unitsquare,p);
```

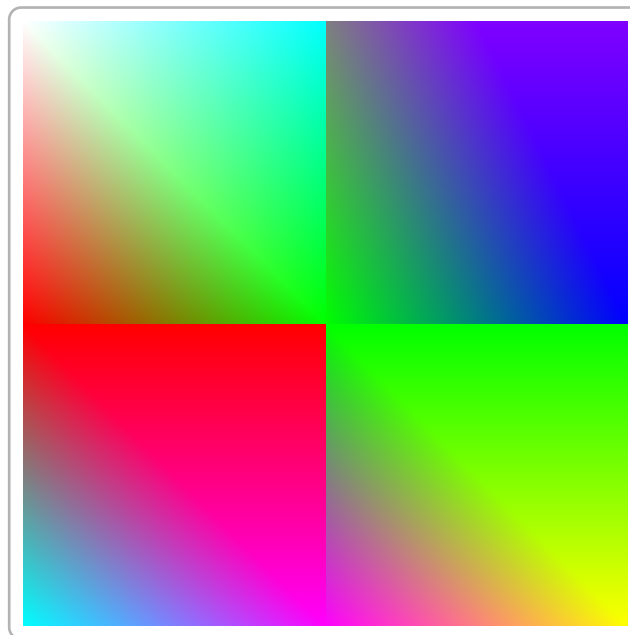


Figure 54: asy-54

```
size(8cm,0);
import palette;
real[][] v={{1,2},{3,4}};
pen[] Palette=Rainbow();
latticeshade(box((0,0),(1,1)),palette(v,Palette));
```

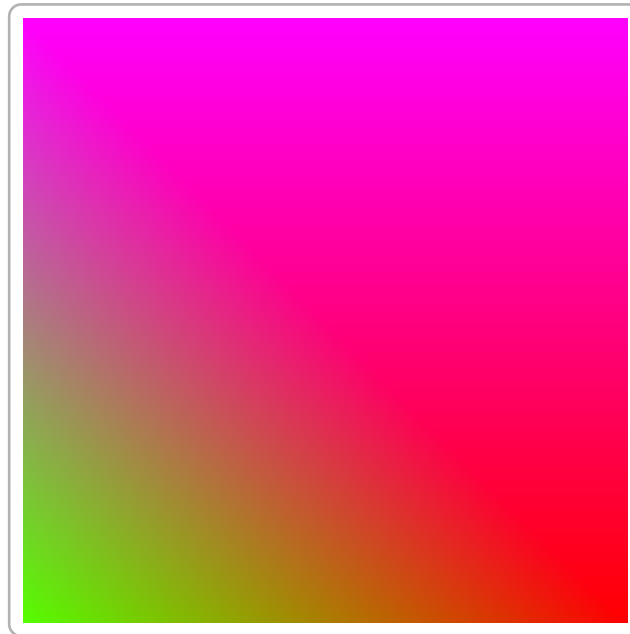


Figure 55: asy-55

```

size(8cm,0);
pair A=(0.35,0.35), B=(0.6,0.6);
radialshade(unitsquare,black,A,0.15,lightgrey,B,0.6);
dot(A,0.8red+8pt);
dot(B,yellow+8pt);
draw(shift(A)*scale(0.15)*unitcircle,dashed+red+1.5pt);
draw(shift(B)*scale(0.6)*unitcircle,dashed+blue+1.5pt);
clip(unitsquare);

```

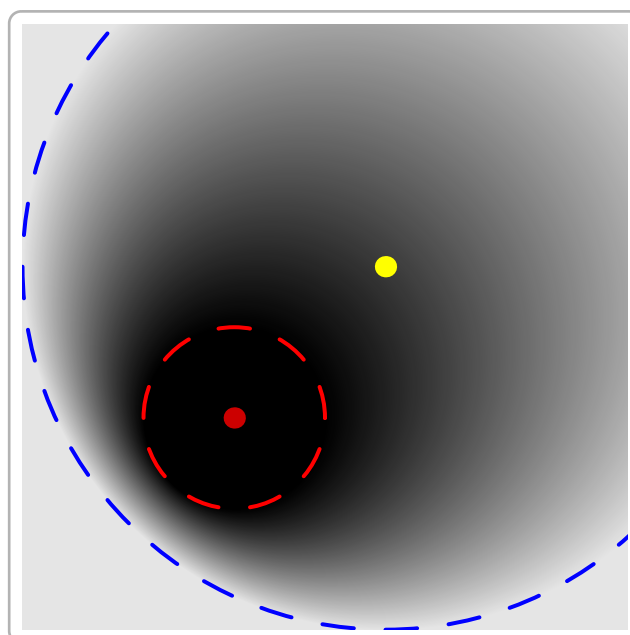


Figure 56: asy-56


```

size(8cm,0);
transform t=xscale(1.25);
pen p1=blue, p2=yellow;
pair pa=t*dir(135), pb=t*dir(-45);
axialshade(t*unitcircle,p1,pa,p2,pb);
draw(pa--pb, dashed);

```

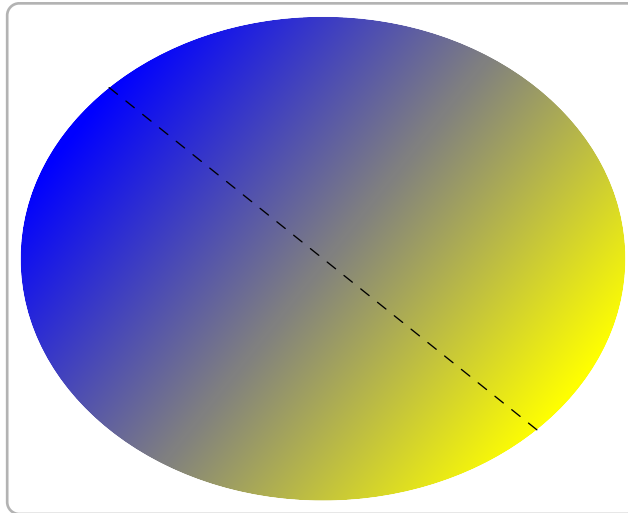


Figure 57: asy-57

```

size(10cm,0);
real r=1;
real R=3.8;
int step=30;
path p=arc(0,r,0,step);
path P=arc(0,R,step,0);
for(int h=0; h < 360; h += step) {
    transform t=rotate(90-h);
    tensorshade(t*p--t*P--cycle,
        new pen[] {white,white,hsv(h-step,1,1),hsv(h,1,1)});
}
for(int h=0; h < 360; h += 30) {
    pair v=R*dir(90-h);
    draw(Label(string(h)+"^\circ",EndPoint),(v--1.05v));
}
draw(circle(0,r));
draw(circle(0,R));

```



Figure 58: asy-58

```
size(16cm,0);
path[] P=texpath("$\displaystyle\int_{-\infty}^{+\infty}
e^{-\alpha x^2}\,,dx=\sqrt{\frac{\pi}{\alpha}}$");
pair m=min(P), M=max(P);
axialshade(P,yellow,m,red,(m.x,M.y));
draw(P,0.5*blue);
shipout(bbox(3mm,Fill));
```

$$\int_{-\infty}^{+\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Figure 59: asy-59

```
size(8cm,0);
pair A, B, C, D;
```

```
A=(0,0); B=(1,0);  
C=(1,1); D=(0,1);  
fill(A--C--B--D--cycle, grey);
```

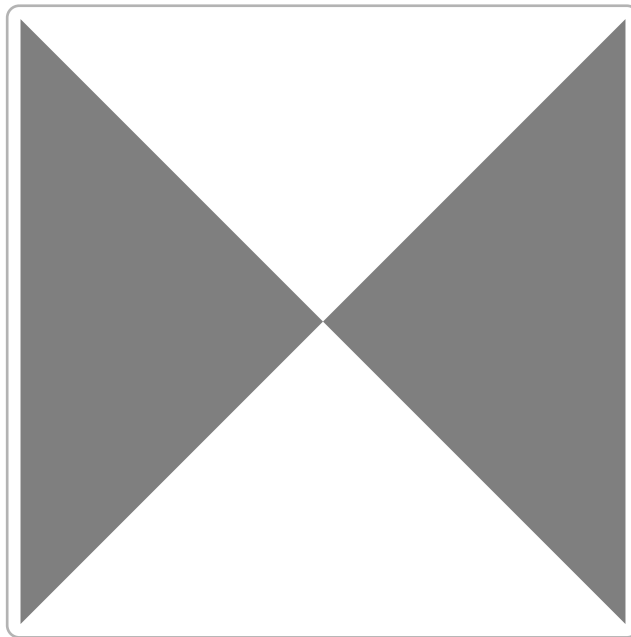


Figure 60: asy-60
